A VARIETY WITH A SMOOTH RATIONAL POINT IS GEOMETRICALLY INTEGRAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the parenthetical '(some details omitted)' at the preservation of density and scheme theoretic density under base change; the parenthetical is reproduced verbatim, and the surrounding reduction and its consequences are written out.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-variety-with-smooth-rational-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A connected scheme with a rational point is geometrically connected. Same chapter, Lemma lemma-geometrically-connected-if-connected-and-point, invoked at proof lines 4788-4789. Supplies geometric connectedness of the smooth locus.
- Smoothness implies geometric normality and reducedness. Same chapter, Lemma lemma-smooth-geometrically-normal, invoked at proof lines 4790-4791. Supplies that the base change of the smooth locus is reduced and normal.
- A connected normal Noetherian scheme is integral. Properties, Lemma properties-lemma-normal-Noetherian, invoked at proof lines 4792-4793. Supplies integrality of the base change, hence geometric integrality of the smooth locus.

Source reference. Original lemma 04KV: sources/stacks-varieties/pinned-original-source.tex, source lines 1082–1082. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 056T: sources/stacks-varieties/pinned-original-source.tex, source lines 4617–4617. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 033M: sources/stacks-chow/tags.txt, source lines 4061–4061. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a variety over k which has a k-rational point x such that X is smooth at x. Then X is geometrically integral over k.*

Proof. Let $U \subset X$ be the smooth locus of X. By assumption U is nonempty and hence dense and scheme theoretically dense. Then $U_{\overline{k}} \subset X_{\overline{k}}$ is dense and scheme theoretically dense as well (some details omitted). Thus it suffices to show that U is geometrically integral. Because U has a k-rational point it is geometrically connected by Lemma 04KV. On the other hand, $U_{\overline{k}}$ is reduced and normal (Lemma 056T. Since a connected normal Noetherian scheme is integral (Properties, Lemma 033M) the proof is complete. □